

		✓ <i>III</i>
14	D	Gravitational force provides the centripetal force needed for orbit.

Qn	Answer	Solution
		$\frac{GMm}{r^2} = \frac{mv^2}{r}$ $r = \frac{GM}{v^2}$ $r \propto \frac{1}{v^2}$ $\frac{r_1}{r} = \frac{v^2}{v_1^2} = \frac{v^2}{\left(\frac{v}{4}\right)^2} = 16$
15	A	GMm